

CredenceAI

Task-backed credit infrastructure for autonomous AI agents.

FinTech | Problem Statement 1: Credit for Autonomous Agents

Verify

Underwrite

Control

Repay

Audit

A safety-first lending layer where AI can request credit, but deterministic controls decide and enforce it.

LINKS

GitHub: <https://github.com/Hardik182005/Fintech-Innova-Hack>

Live Demo: <https://credence-web-ppmafqinnq-as.a.run.app>

Team Kryptonite

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Autonomous agents can spend, but the financial system cannot yet lend to them safely.

- Agents may need short-term working capital for compute, APIs, data or vendor payments before task revenue arrives.
- Existing lending assumes a legal borrower who can sign, pledge collateral and be chased for repayment.
- An AI agent has no standalone legal identity, no traditional credit score and no direct collection mechanism.
- The gap blocks useful agent work even when the task is valuable and revenue is expected.

What lenders need to see

- Who authorized the agent
- Why this task is worth financing
- How repayment will be recovered
- How downside is capped if it fails

Our interpretation

This is not a chatbot credit score. It is controlled credit issued against a specific task, with repayment and spending limits enforced by software.

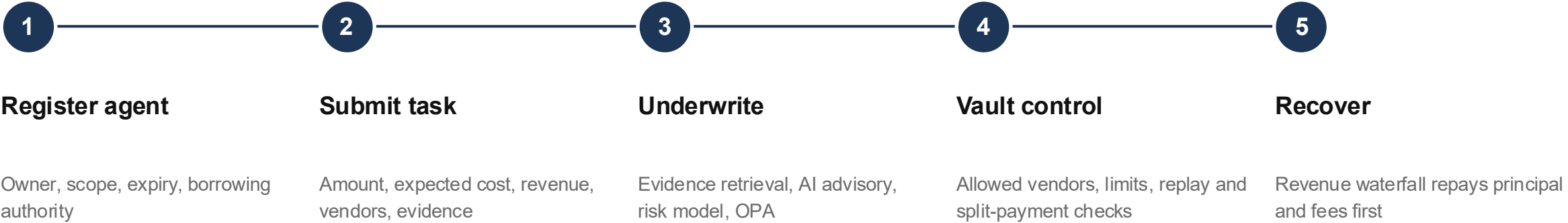
CredenceAI gives an agent credit for a task - not unrestricted money.

- Agent Passport links the AI agent to a verified owner, task scope, authorization window and borrowing limit.
- Underwriting uses behavioral history: task success, repayment record, spending pattern, policy violations and expected task margin.
- Approved credit goes into a restricted vault that can pay only allowed vendors within policy limits.
- Revenue is collected into a repayment waterfall before any remaining proceeds go to the owner.
- If anything is missing or suspicious, the system fails closed into rejection or human review.

The operating principle

AI recommends. Deterministic systems decide. Financial controls enforce.

From credit request to repayment, every stage leaves evidence.



Repayment equation

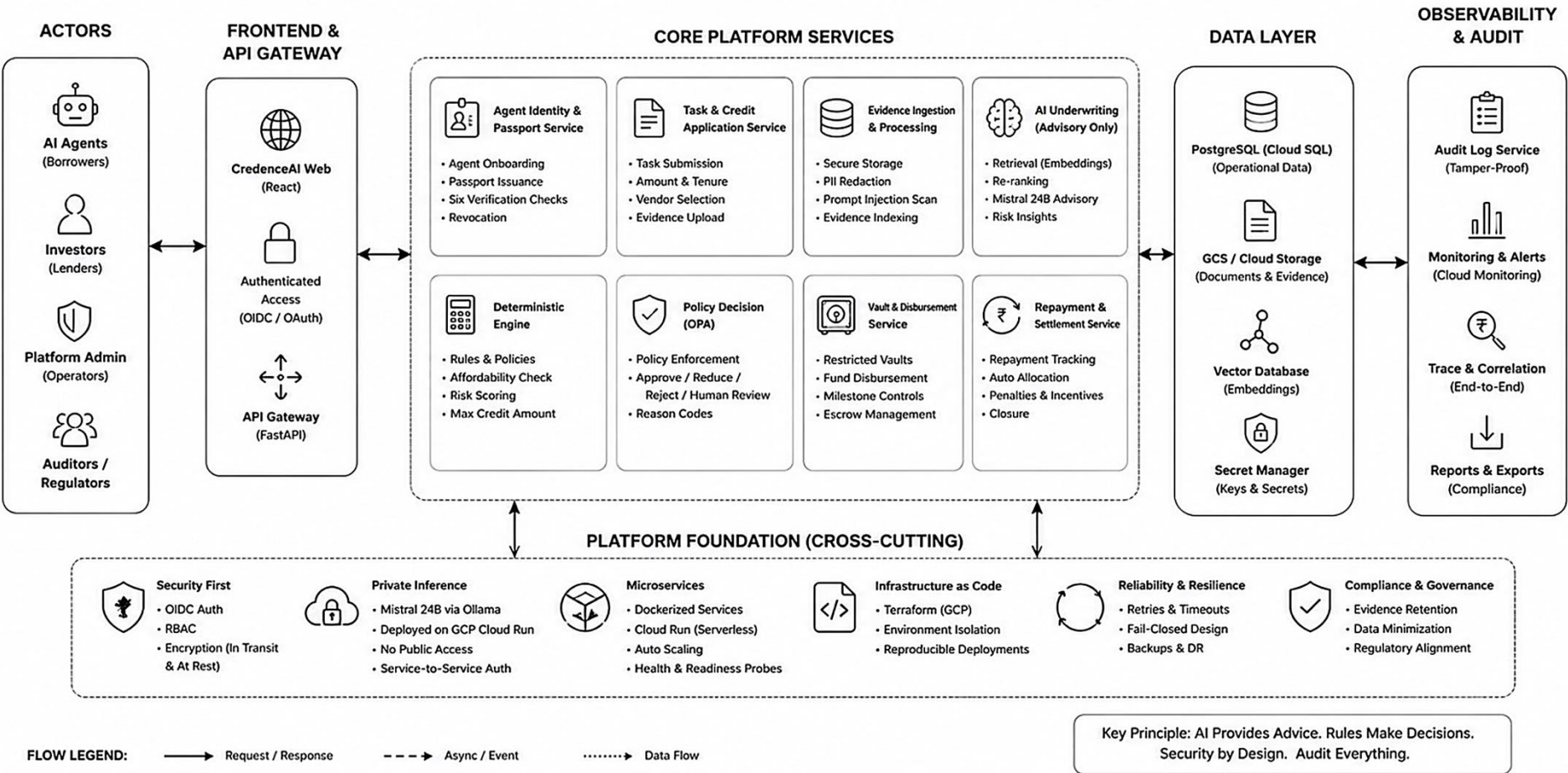
Task revenue = principal recovered + credit fee + platform fee + owner proceeds

Why it matters

- The agent never receives open-ended cash.
- The lender is repaid before proceeds are released.
- Every event is recorded in an audit chain.

CredenceAI Architecture

Agentic AI for Task-Backed Credit with Deterministic Control, Private Inference & Full Auditability



The live web app works as a finance control room for agent credit.

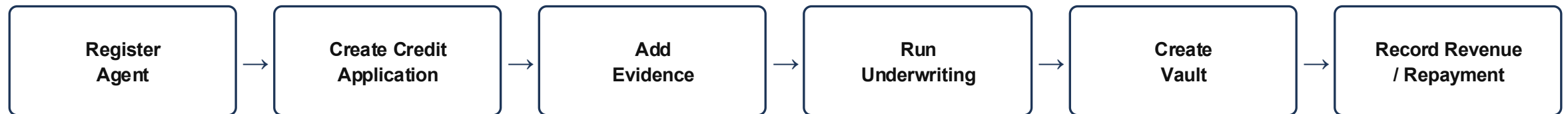
Operator pages

- Overview: approved credit, repayment rate, blocked exposure and ledger health
- AI Agents: register agents, view status, owner, trust score and passport
- Credit Applications: enter task, amount, evidence and expected revenue
- Underwriting: compare AI advisory, deterministic limit and OPA result
- Vaults, Transactions and Repayments: enforce spend and recovery

Judge-facing proof

- Risk Monitoring: overspend, unknown vendor, split-payment and replay blocks
- Audit Trail: tamper-evident sequence of key financial events
- System Intelligence: model versions, retrieval metrics, GPU health and fail-closed events
- Judge Demo: one-click scenarios for happy path and adverse cases

Core input flow



The model helps with judgement; it does not control funds.

Local model stack

- Mistral 24B on a private NVIDIA L4 service for advisory underwriting
- Qwen embedding and reranker for tenant-scoped evidence retrieval
- Guard model plus Presidio for prompt-injection and PII handling
- PostgreSQL with pgvector keeps evidence close to relational permissions

Financial authority

- Logistic/XGBoost-style risk model plus deterministic scorecard
- OPA enforces identity, limits, vendors, replay, expiry and revocation
- Vault service enforces spend; repayment service enforces recovery
- Model failure returns human review or analysis unavailable - never approval

No direct external LLM underwriting. No private keys, wallet credentials or database passwords are given to the model.

Each core feature maps directly to the judging criteria.

Trust design	Agent Passport, owner authorization, evidence IDs, behavioral history
Repayment enforceability	Restricted vault, escrow-style revenue flow, exact repayment waterfall
Risk containment	Credit limits, allowlists, split-payment detection, kill switch, fail-closed paths
Technical soundness	Live GCP sandbox, private API/inference, Cloud SQL, OPA, audit chain
Real-world plausibility	Private local models, tenant isolation, idempotency, observability and cost controls

The judge should see a working system, not just a concept: new input, live decision, blocked attack, repayment, audit and System Intelligence evidence.

- Every credit line is tied to a specific task, a verified owner and an enforced spending limit — never open-ended cash.
- Restricted vaults and a fixed repayment waterfall recover principal and fees before any owner proceeds are released.
- Deterministic policy and risk engines make every approval decision; the AI model only advises and can never approve.
- A full audit trail, kill switch and fail-closed safeguards contain risk at every stage, from request to repayment.

CredenceAI makes autonomous-agent credit measurable, revocable and recoverable.

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